

SAFE OPERATING PROCEDURE

Oxy-Acetylene Gas Set LOCATION: Various

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PERSONAL PROTECTIVE EQUIPMENT



Version: 2	Key Contact: George Fisher
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Permits required:N/A	Confined space ☐ Hot Works ☐ Working at Heights ☐

HAZARDS ASSOCIATED WITH THIS EQUIPMENT

PERMANENT HAZARDS Remove as necessary	List	Control Method	Lock out Yes or No
Fire & Explosion	Sparks can be produced from the flame and metal contact, gas leakage can cause explosion, gases are combustible	Keep all flammable material away from the area, check cylinders and hoses are in good repair and no leaks, read SDS sheets for gases	No
Burns	From gas flame	Wear gloves and non-flammable clothing	No
Eye injury	Radiation can damage eyes and flying sparks may enter eyes	Wear welding helmet/welding goggles and ensure it is fitted with appropriate filter	No
Hot surfaces	Can cause burns	Gloves	No

PURPOSE/PROCESS

PROCEDURES:

DO NOT use this machine unless you have proven to your trainer that you are competent in its safe use and operation

Pre Start Check

1. Ensure no slip/trip hazards are present in the workspace and make sure bottles are secured (chained) before commencing work
2. Keep work area clean and free of grease, oil and other flammable material.
3. DO NOT lubricate any hose, hand piece, valve or other connections with grease or oil.
4. Gas hoses, flashback arrestors, regulators and hand pieces must be visually inspected and found to be in good condition before using the equipment.

5. When changing oxy-acetylene bottles all lines and connections must be checked for leaks using appropriate method (soapy water) and recommended spanner used.
6. Check that the area is well ventilated before using any welding equipment
DO NOT use faulty equipment – immediately report suspect machinery.

Pressure Setting

1. Check that the oxygen and acetylene/fuel regulator adjusting knobs are loose.
2. Check that both hand piece valves are closed.
3. Slowly open the cylinder valves on each cylinder for half a turn only.
4. Screw in the regulator adjusting knobs slowly until the delivery pressure gauges register 70kPa.
5. Purge and check for constant oxygen gas flow.
 - a. Open the oxygen line valve for 2 seconds and check the delivery gauge.
 - b. If necessary re-adjust the oxygen regulator to achieve a 70kPa pressure
 - c. Close the oxygen line valve
6. Purge and check for constant acetylene/fuel gas flow
 - a. Open the acetylene/fuel line valve for 2 seconds and check the delivery gauge.
 - b. If necessary re-adjust the acetylene regulator to achieve a 70kPa pressure.
 - c. Close the acetylene line valve.



Lighting Up

1. Open the acetylene/fuel blowpipe valve slightly and light the blowpipe with a flint lighter.
2. Continue to slowly open the acetylene/fuel valve until the flame no longer produces soot.
3. Slowly open the oxygen blowpipe valve until the required flame is produced.
4. Beware of blowback from being too close to work.

Shutting off gas set

1. Close the acetylene/fuel valve first.
2. Then close the oxygen valve.

Closing down

1. Close down both cylinder valves first.
2. Open oxygen valve to allow the line to vent.
3. When oxygen gauges read zero, unscrew regulator-adjusting knob.
4. Close oxygen valve.
5. Turn off acetylene/fuel cylinder valve.
6. Open acetylene/fuel valve and release gas.
7. When acetylene/fuel gauges read zero, release regulator adjusting knob.
8. Close acetylene/fuel valve.

Housekeeping

1. Hang up welding hand piece and hoses and clean area.

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2. Leave the work area in a safe, clean and tidy condition.

Forbidden:

1. Lighting the hand piece with matches or lighters
2. Using oil, greases or other hydrocarbons near or around Oxy/acetylene gases.
3. Using oxygen as a substitute for compressed air.

	Name	Title/Position	Signature	Date
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Manager	Joe Nathan			

Document Control

Date	Revision number	Modifications	Revised by	Approved by
28/11/2019	1	New Template and photos	George Fisher	Marie Dennis
17/03/2020	2	Updated	George Fisher	M Dennis